

Meta-analysis of Empirical Estimates of Loss Aversion[†]

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Loss aversion is one of the most widely used concepts in behavioral economics. We conduct a large-scale, interdisciplinary meta-analysis to systematically accumulate knowledge from numerous empirical estimates of the loss aversion coefficient reported from 1992 to 2017. We examine 607 empirical estimates of loss aversion from 150 articles in economics, psychology, neuroscience, and several other disciplines. Our analysis indicates that the mean loss aversion coefficient is 1.955 with a 95 percent probability that the true value falls in the interval [1.820, 2.102]. We record several observable characteristics of the study designs. Few characteristics are substantially correlated with differences in the mean estimates. (JEL D81, D91)

1. Introduction

Loss aversion is the empirical observation that decisions often reflect a larger distaste for potential losses, compared to equal-sized gains, relative to a point of reference. Loss aversion is a core feature

of prospect theory (Kahneman and Tversky 1979, Tversky and Kahneman 1992, Wakker 2010), an explicitly descriptive model of choice under risk and uncertainty that has been widely applied and cited.¹ The strength of aversion to loss compared to attraction to

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¹The 1979 paper is the most widely cited empirical economics paper published from 1970–2005 (see table 2 in Kim, Morse, and Zingales 2006). Note also that Fishburn and Kochenberger (1979) documented loss aversion in a different sample of preferences elicited for decision analysis.

gain is typically captured by a single parameter, λ .

In his popular-science book, Kahneman (2011) writes that “the concept of loss aversion is certainly the most significant contribution of psychology to behavioral economics” (p. 300). Loss aversion has been widely applied to many types of economic decisions and analyses. It is often applied in analyses of experimental decisions over monetary risks (as in the original Kahneman and Tversky 1979). However, the use of loss aversion and dependence on reference points has evolved well beyond its initial application. Applications include financial asset prices (Barberis 2013), the equity premium puzzle (Benartzi and Thaler 1995), labor supply decisions (Camerer et al. 1997), political power of entitlements changes (Romer 1996), majority voting and politics (Alesina and Passarelli 2019), sectoral trade policy behavior (Tovar 2009), and selling–buying price endowment effects in contingent valuation of nontraded goods (Ericson and Fuster 2014, Tunçel and Hammitt 2014). Loss aversion also features prominently in behavioral industrial organization, in theories and evidence of responses to price changes (Heidhues and Kőszegi 2018).

Several different methods have been used to measure loss aversion. These include laboratory experiments, representative panel surveys, analyses of natural data, and randomized trials trying to change behavior. Loss aversion has been quantified for monetary outcomes as well as for nonmonetary outcomes such as health (Attema, Brouwer, and l’Haridon 2013). Other fields outside economics also utilize loss aversion, including neuroscience, psychiatry, business and management, and transportation.

Given how widely the concept of loss aversion has been applied in economics and many other social sciences, it is useful to have the best possible empirical answer about how large loss aversion is and how it

varies. One of the first empirical estimates of λ is reported in Tversky and Kahneman (1992). The authors elicit the preferences of 25 graduate students from elite west-coast American universities using three sessions of unincentivized lottery choice experiments. The median λ —no mean nor statistic of dispersion was reported—was $\lambda = 2.25$ (figure 1). Many analyses, to this day, cite this number as the typical degree of loss aversion. For example, the value $\lambda = 2.25$ is used in numerical simulations of prospect theory in behavioral finance (e.g., Barberis, Huang, and Santos 2001; Barberis and Huang 2001, 2008; Barberis and Xiong 2009; Barberis, Mukherjee, and Wang 2016; Barberis, Jin, and Wang 2021). Of course, had Tversky and Kahneman initially reported a different value (e.g., 1.5) these analyses might yield different findings. Some of these authors are well aware of this issue.

As noted in the latter study, “[...] these estimates are almost 30 years old and are based on a small number of participants. Given that the values we assign to these parameters play a significant role in our results, it seems prudent to base these values on a wide range of studies, not just one.” (Barberis, Jin, and Wang 2021, p. 2665).²

What is the best way to cumulate knowledge about λ after thirty years of research?

²It is not our contention that there is a consensus in the academic community around the estimate $\lambda = 2.25$, though there may be some anchoring to that value (as we have just described). Our general contention is that there is no consensus and a lack of confidence in a uniform estimate. In fact, during one of the early presentations of this paper at the Economic Science Association World Meeting in Vancouver in 2019, we elicited guesses of our mean parameter. We incentivized the audience to guess correctly with a CA\$50 dollar prize for the closest guess. We have 37 guesses, and 34 participants also reported their confidence levels (low, medium, or high). Mean guesses (of the mean parameter) were 1.639 with a standard deviation of 0.599. Of the 34 answers, 20 (58.8 percent) reported low confidence in their guesses and nine (24.3 percent) fell between 1.8 and 2.1. See online appendix F for the full distribution of the guesses.

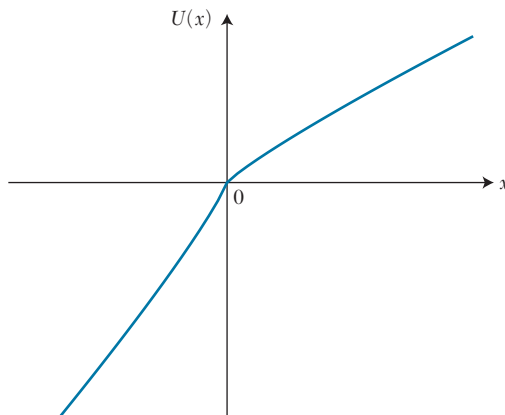


Figure 1. An Example of Prospect Theory Utility Function

Notes: This is specification (3) presented in section 2, $U(x) = x^\alpha$ for $x \geq 0$ and $U(x) = -\lambda(-x)^\beta$ for $x < 0$, with median parameters of $\lambda = 2.25$ and $\alpha = \beta = 0.88$ reported in Tversky and Kahneman (1992).

Our view is that *meta-analysis* is an indispensable tool for scientific cumulation. Meta-analysis is a principled, reproducible, open-science method for accumulating scientific knowledge (and also for detecting nonrandom selective reporting of evidence: Stanley 2001, Stanley and Doucouliagos 2012). A meta-analysis uses a clearly specified method of sampling available studies, coding evidence in a way that is comparable across studies, and summarizing both regularity and variation across studies. The idea of synthesizing evidence from multiple studies dates back to the early 1900s (Pearson 1904, Yates and Cochran 1938), but the history of modern meta-analysis has its origin in the 1976 American Educational Research Association (AERA) presidential address by Gene V. Glass. He introduced the term “meta-analysis” to refer to “the statistical analysis of a large collection of analysis results from individual studies for the purpose of integrating the findings” (Glass 1976, p. 3). It has been widely used in evidence-based practices in medicine and

policy for at least two decades (Gurevitch et al. 2018). However, meta-analysis has been mostly absent from highly selective journals in empirical economics.³

This paper reports the results of a meta-analysis of empirical estimates of loss aversion. The dataset comprises 607 estimates reported in 150 papers in economics, psychology, neuroscience, and several other disciplines.

³Prominent meta-analyses in economics include value of a statistical life (Doucouliagos, Stanley, and Giles 2012; Doucouliagos, Stanley, and Viscusi 2014), intertemporal elasticity (Havránek 2015, Havránek et al. 2015), habit formation (Havránek, Rusnak, and Sokolova 2017), foreign direct investment (Iršová and Havránek 2013), minimum wage effects (Card and Krueger 1995, Doucouliagos and Stanley 2009), gender wage discrimination (Stanley and Jarrell 1998), microcredit interventions (Meager 2022), behavior in dictator and ultimatum games (Engel 2011; Oosterbeek, Sloof, and van de Kuilen 2004), preferences for truth telling (Abeler, Nosenzo, and Raymond 2019), experimentally measured discount rates (Matoušek, Havránek, and Iršová 2022), and present bias in Convex Time Budget experiments (Imai, Rutter, and Camerer 2021).

The tool kit of meta-analysis can give the best available answers to three questions:

- (i) What is the central tendency in the distribution of λ estimates and how much do they vary?
- (ii) Does measured λ vary systematically across different methods, definition of λ , utility specifications, domains of choice, and types of participants?
- (iii) Is there evidence of selective reporting, or publication bias, which distorts reported estimates of λ compared to the corpus of ideal evidence without such biases?

While the answers to these questions no doubt carry some intrinsic interest to researchers interested in loss aversion, they also have practical validity. For one, this paper's mean estimate of $\lambda = 1.955$ provides a much more informed value of loss aversion than the original $\lambda = 2.25$ for researchers to use as an input in financial models (see above). The results can help researchers do their work better in several other ways.

Imagine a researcher who is interested in loss aversion but not quite sure what steps to take to measure it or to apply it. First, the researcher might ask: What method should I use to measure λ ? What are the most popular methods? Does it make much difference which one is used? Results on how estimated λ 's vary with characteristics of the measurement method, such as the type of the data (experimental or field), reward (monetary or non-monetary), specification of the utility function, and the definition of loss aversion, can guide the researcher.⁴

Second, the researcher may be doing a behavior change intervention leveraging the psychology of loss aversion. Then she needs a specific estimate of λ or a plausible range of values to use to make a power calculation. Perhaps she is planning to prepay teacher bonuses, which they can later lose, to motivate them to increase student test outcomes (Fryer et al. 2012). Is $\lambda = 2.25$ a good guess or is there a better guess? Is there a more refined estimate of λ for the subset of studies in the meta-analysis that are most like the one she is planning? Meta-analysis can help here too.

Third, suppose the researcher has just read review articles about prospect theory and reference-dependent preferences (e.g., Barberis 2013; DellaVigna 2009, 2018; O'Donoghue and Sprenger 2018). Those reviews have a "narrative" programmatic structure in which results of early and key studies raise fundamental questions that later studies are designed to answer. The reader is usually left with an understanding of the historical intellectual trajectory and what the next wave of studies should try to understand better. The researcher wonders, is anything important left out of the narrative? The meta-analysis helps answer this question too. (However, the comparison and complementarities of meta-analysis and narrative review are subtle and important, so we will return to them in the conclusion.)

Figure 2 shows the distribution of loss aversion coefficients λ in our dataset, where the median value of the raw data points is 1.69 and the mean is 1.97. The distribution is skewed right and has a substantial mass (93.9 percent) on the range $\lambda > 1$, corresponding to loss aversion (as opposed to loss tolerance, $\lambda < 1$). Applying a Bayesian hierarchical approach taking into account the uncertainty surrounding the measurements,

⁴These methodological variations can also help us understand the mechanisms behind loss aversion, along

with process measures such as response times, psychophysiology, and neuroscientific data.

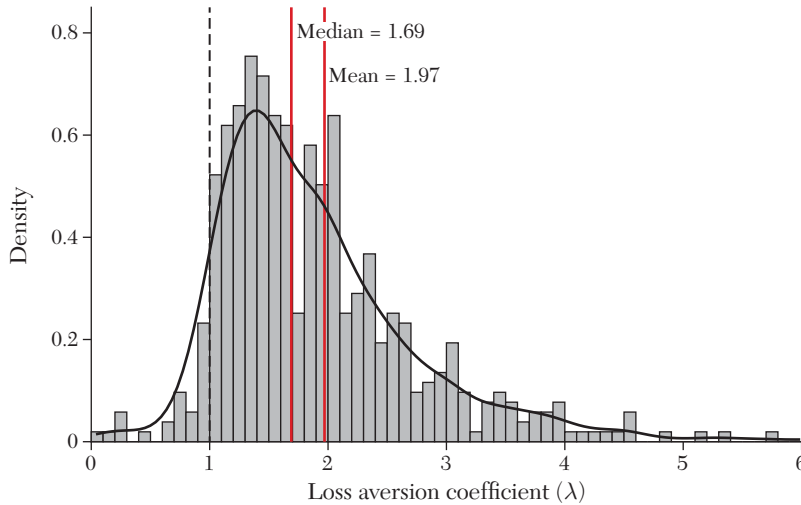


Figure 2. Distribution of Reported Estimates of Loss Aversion Coefficient

Notes: Bins for the histogram are 0.1 wide. The kernel density estimate of the distribution is superimposed, using the Gaussian kernel with Silverman's rule of thumb for the bandwidth selection. There are 85 cases that report both individual-level mean and median. We keep individual-level medians from these cases. The x-axis is cut off at 6 for better visual rendering, but the density estimation keeps five observations with $\lambda > 6$.

we find that the average λ in the literature lies between 1.7 and 1.9. Taking into account the fact that many papers reported more than one estimate (thus producing correlation among estimates), the average is between 1.8 and 2.1. We also examine whether observed heterogeneity in reported λ can be attributed to some of the observable characteristics of the study design. The results do not show many strong reliable effects.

Even to economists unfamiliar with meta-analysis, the method should be, in some ways, familiar. It is essentially an application of econometric techniques to literature review (see Stanley and Doucouliagos 2012). Like for any empirical study, the greatest concern should be the inclusion criteria of the dataset (i.e., selection). While our broad inclusion criteria are independent

of estimates of λ , we have little control over publication decisions (which makes papers more prominent and easier to find) and whether a study is written at all (i.e., “the file drawer problem”; Rosenthal 1979), which could be dependent on the values of estimated λ . We consider these issues and how they might affect our analysis by examining the correlation between estimated λ and their standard errors and by inspecting the shape of distributions of z -scores. As noted elsewhere, no technique, not even the alternative narrative review, can address this issue perfectly (Borenstein et al. 2009). Meta-analysis at least has the tools to examine these possible issues quantitatively.

Finally, we note an advantage of meta-analysis is that as new evidence arrives, it can be easily added to the previous corpus of studies and results can be quickly updated.

1.1 Related Papers

There are two previous meta-analyses of loss aversion, to which we contribute a newer and broader scope.⁵ Neumann and Böckenholt (2014) conducted a meta-analysis of 109 estimates of loss aversion from 33 studies about consumer brand choice. As we do later in this paper, they use a multi-level, random-effects technique to account for the variability of estimates, both within and between studies, of the logged- λ parameter. They report a base model estimate of $\lambda = 1.49$ and an “enhanced model” estimate of $\lambda = 1.73$, accounting for sources of estimate variability. Perhaps because of their narrow focus on consumer choice, their meta-regression controls explain nearly all of the variability within their data. Notably, the use of external versus internal reference points, estimates derived from models that account for both heterogeneity in taste and process, and unpublished versus published studies are all associated with lower estimates of loss aversion.

A different approach was used by Walasek, Mullett, and Stewart (2018) to understand heterogeneity in a narrow domain of mixed gain-loss financial lotteries. Their analysis used only published experimental studies of mixed lotteries of gains and losses where original raw data were available for reanalysis. Their corpus is 19 estimates from 17 articles.⁶ Rather than meta-analyzing estimates from the original papers, they reestimated parameters for a single model of cumulative prospect theory (i.e., power utility function

with symmetric curvature, $\alpha = \beta$, see equation (3) in section 2) using the original data. Their random-effects meta-analysis on the 19 estimates has an average $\lambda = 1.31$. Despite their rather strict restrictions, the authors note that there are high levels of methodological variability between studies (their data is not very useful in looking at this question within studies) in both estimates and procedures.

The rest of the paper is organized as follows. Section 2 introduces the concept of loss aversion in prospect theory. Section 3 describes how we assembled the dataset of empirical estimates of loss aversion. Section 4 provides results, and section 5 discusses their implications.

2. Loss Aversion

In this section, we briefly illustrate some typical definitions of loss aversion in prospect theory. Consider a situation where an agent makes a choice under risk between prospects with at most two distinct outcomes. This simplified structure still captures a wide range of empirical studies examined here. Let $(x, p; y)$ denote a *simple lottery*, which gives outcome x with probability p and outcome y with probability $1 - p$ (Abdellaoui, Bleichrodt, and Paraschiv 2007; Chateauneuf and Wakker 1999; Köbberling and Wakker 2005). A key assumption of prospect theory is that outcomes are evaluated as *gains* and *losses* relative to a *reference point*. For simplicity of exposition, in this section, we assume the reference point to be 0 so that the sign of the outcome indicates whether it is a gain or a loss. We call a lottery *non-mixed* if two outcomes have the same sign (i.e., either $x, y \geq 0$ or $x, y \leq 0$) and *mixed* if one of the outcomes is positive and the other outcome is negative. Without loss of generality, we assume that $x > 0 > y$ when we deal with a mixed lottery.

⁵Mrkva et al. (2020) ask a question similar to ours, examining how individual differences moderate the degree of loss aversion. Their approach is different—they conduct five large-scale field surveys with a total of 17,720 subjects.

⁶Though not specifically excluded by the aforementioned criteria, the authors also excluded studies that relied on adaptive questions because of concerns about how such techniques would affect their maximum likelihood estimation procedures (i.e., Abdellaoui, Bleichrodt, and l'Haridon 2008; Wakker and Deneffe 1996).

In this setup, both the original prospect theory by Kahneman and Tversky (1979) (hereafter OPT) and its modern incarnation, the cumulative prospect theory of Tversky and Kahneman (1992) (hereafter PT), postulate that the agent evaluates non-mixed prospects $(x, p; y)$ with $x \geq y \geq 0$ or $x \leq y \leq 0$ by

$$(1) \quad w^s(p)U(x) + [1 - w^s(p)]U(y),$$

and mixed prospects $(x, p; y)$ with $x > 0 > y$ by

$$(2) \quad w^+(p)U(x) + w^-(1 - p)U(y),$$

where $w^s : [0, 1] \rightarrow [0, 1]$ is a probability weighting function for gains ($s = +$) or for losses ($s = -$), with $w^s(0) = 0$ and $w^s(1) = 1$, and $U : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing utility function satisfying $U(0) = 0$. Tversky and Kahneman (1992) assume that the utility function U and the probability weighting functions w^+ and w^- exhibit diminishing sensitivity. Note also that $w^+ = w^-$ is assumed under OPT, and expected utility is a special case of PT where $w^+(p) = w^-(p) = p$ for all $p \in [0, 1]$.

A particularly popular functional apparatus is the one using different power utility parameters for gains and losses, following the approach of Tversky and Kahneman (1992):

$$(3) \quad U(x) = \begin{cases} x^\alpha, & \text{if } x \geq 0; \\ -\lambda(-x)^\beta, & \text{if } x < 0; \end{cases}$$

where $\lambda > 0$ is the *loss aversion coefficient*, the target variable of interest in this study. Values of $\lambda > 1$ are taken to indicate loss aversion, whereas values of $\lambda < 1$ indicate loss tolerance (also referred to as “gain seeking”), with $\lambda = 1$ indicating loss neutrality. The utility function is concave for gains and convex for losses, reflecting diminishing sensitivity, when $\alpha, \beta \in (0, 1)$.

Notice that mixed prospects are necessary to identify loss aversion, since λ cancels out in the evaluation of pure loss prospects such as in equation (1). In this particular formulation, the loss aversion parameter is dependent on the scale of the data, and thus not uniquely defined due to scaling issues (see Wakker 2010, section 9.6, for a theoretical discussion). If, on the other hand, the two power parameters are assumed to be identical, that is, $\alpha = \beta$, this issue does not occur. It also does not occur for different utility parameters using alternative functional forms, such as exponential utility (Köbberling and Wakker 2005).

Beyond the popularity of the formulation provided above, it is important to note that several different definitions have been proposed and used in the literature. Köbberling and Wakker (2005) and Abdellaoui, Bleichrodt, and Paraschiv (2007) provide extensive discussions of such alternative definitions, which we summarize in online appendix B. Furthermore, under PT decision weights (given by probability weighting functions w^+ and w^-) naturally enter the definition of loss aversion (Schmidt and Zank 2005). The combination of different definitions with different functional forms for utility and weighting functions results in a large variety of different formulations. Our strategy in this meta-analysis is simply to take the estimate emerging from the formulation of the authors. We code the type of definition adopted to be able to determine the correlation of definitions and functional forms with estimates.

A definition that more clearly departs from the apparatus presented above is the expectation-based reference-dependent model of Kőszegi and Rabin (2006, 2007). In this model, an agent evaluates a consumption outcome x by

$$v(x|r) = m(x) + \mu(m(x) - m(r)),$$

where the function m represents the direct utility from consumption and the function μ represents the gain–loss utility from departures from a reference point r . In typical applications of the model the consumption utility m is assumed to be linear, so that $m(x) = x$, and a piecewise linear gain–loss utility function is adopted:

$$\mu(z) = \begin{cases} \eta z, & \text{if } z \geq 0; \\ \lambda \eta z, & \text{if } z < 0; \end{cases}$$

where the parameter $\eta \geq 0$ captures the importance of the gain–loss utility relative to the consumption utility, and λ again captures loss aversion.⁷

3. Data

3.1 Identification and Selection of Relevant Studies

In order to deliver an unbiased meta-analysis, we first identified and selected relevant papers following unambiguously specified inclusion criteria. The main criterion is to include “all empirical papers that estimate a coefficient of loss aversion.” Note that, under this criterion, we include papers that use choice data from laboratory or field experiments and also nonexperimental, naturally occurring data including stock prices, TV game shows, and surveys on transportation.

We searched for relevant papers on the scientific citation indexing database Web of Science. The initial search, made in the summer of 2017, returned a total hits of 1,547 papers. As a first step of paper identification, we went through titles and abstracts and threw out 833 papers that were irrelevant to our study. We then read the remaining

papers, applied our inclusion criteria based on the content, and coded information (described in section 3.2 below). Finally, we posted a message on the email list of the Economic Science Association to ask for relevant papers (in February 2018).

In the initial search phase, we cast a net also to identify papers that estimate the degree of loss aversion in riskless choice, through measuring the discrepancy between willingness to pay (WTP) and willingness to accept (WTA). However, in reading many of these papers, we discovered that while the authors had measured WTP and WTA for a given item, they had not intended to do so as a way to estimate the loss aversion coefficient. While it is straightforward to impose a certain linear utility structure on such papers to recover a loss aversion coefficient, we viewed it as problematic to impose our own assumptions on the work of others, and thus, faced with no better option, did not include these papers.

The search and selection procedure is summarized in online appendix A.1. We identified 150 papers at the end of this process. Twenty papers are unpublished at the time of the initial data collection (summer 2017).

3.2 Data Construction

We assembled the dataset for our meta-analysis by coding relevant information—estimates of the loss aversion coefficient and the associated standard error, characteristics of the data, and measurement methods (Brown et al. 2024). The primary variables of interest are estimates of the loss aversion coefficient λ . These estimates come in two different forms: (i) *aggregate-level*, where a single λ for the “representative” agent/subject is estimated by pooling data from all subjects in a study; and (ii) *individual-level*, where λ is estimated for each subject in a study and the summary statistics of empirical distribution,

⁷For small- to modest-scale risks, consumption utility can be taken to be approximately linear (Kőszegi and Rabin 2007; Rabin 2000).

typically mean or median, are reported. We have a dummy variable capturing the type of reported estimates. We also coded standard errors (SEs) of parameter estimates as a measure of the estimate's uncertainty/precision and the study's quality. All conventional meta-analyses require SEs to calculate weighted averages and to correct for the heteroskedasticity of meta-regression. Other measures of study quality, when known, can be easily included in these weights. When SEs are not reported, we reconstructed them from other available information such as standard deviation (SD), p -value (of the null hypothesis of loss neutrality), or the interquartile range (IQR).⁸

We also coded variables describing the characteristics of the data and measurement methods. These variables include: type of the data (e.g., experimental, nonexperimental, TV game show); location of the experiment (e.g., laboratory, field, online); types of reward (e.g., real or hypothetical, money, health, time); subject population (e.g., children, college students, general population, farmers); definition of loss aversion coefficient (as described in online appendix B); utility specification (e.g., constant relative risk aversion, hereafter CRRA, or constant absolute risk aversion, hereafter CARA); and several others. Table A.1 in the online appendix lists all variables coded in the study.

The set of papers we included spans a wide range of disciplines (see table A.3 in the online appendix). Since fields/journals have different reporting cultures and standards, we could not always retrieve all the

necessary information from reading papers. We thus emailed the authors of the papers when some essential summary statistics of the loss aversion coefficient or sample size information was missing.⁹

While study quality is an ongoing concern for meta-analysis in general (and one that we will revisit in our concluding section), we record quantitative values, such as each estimate's precision and the impact factor of the journal in which the study was published, as measures of study quality in line with past precedent (Stanley and Doucouliagos 2012). Meta-analyses in economics routinely code several dimensions of research quality such as whether the study accounts for endogeneity, is experimental, uses panel data, et cetera. These factors are then included in a meta-regression to estimate the effects of study quality differences and to isolate the findings from higher-quality studies. Here, we recognize quality differences by coding the impact factor of the journal where the study is published, the experimental nature of the study, the subject pool, and the type of reward used.

3.3 Descriptive Statistics

We identified 150 articles that report an estimate of the loss aversion coefficient λ . See online appendix G for the full list of articles included in our meta-analysis. Among these, 130 articles were published in 78 journal outlets (including eight articles published in the "top five" journals in economics). The dataset includes papers from a variety of disciplines: economics, management, psychology, neuroscience, medicine, psychiatry, agriculture, environment, transportation,

⁸We calculated/approximated 68 SEs from other available information: 64 from IQR and sample size and four from p -values. Use of the IQR to infer the SE of the mean (from an approximation of the standard deviation, $SD \approx 1.35 \times IQR$) is in principle only legitimate if the parameters are normally distributed in the population. That assumption is clearly a stretch. Nevertheless, obtaining even an "approximate" SE seemed preferable to dropping the observation entirely, or to making other, even stronger assumptions allowing us to keep the observation.

⁹We contacted the authors of 51 papers asking for additional clarification on 175 estimates (i.e., mean, SE, or the number of observations). If the authors did not respond to our initial request, we sent an additional email. Overall, we received 39 responses. Of those, 28 responses were ultimately useful and we recovered additional information on 78 estimates. The remainder had to be imputed.

TABLE 1
STUDY CHARACTERISTICS

	Frequency	Share (%)		Frequency	Share (%)
<i>Total number of studies</i>	185	100.0			
<i>Data type</i>			<i>Reward type</i>		
Lab experiment	98	53.0	Money	154	83.2
Field experiment	29	15.7	Other	14	7.6
Other field data	20	10.8	Consumption good	8	4.3
Classroom experiment	18	9.7	Mixed	5	2.7
Online experiment	17	9.2	Health	2	1.1
Game show	3	1.6	Food	1	0.5
			Environment	1	0.5
<i>Subject population</i>			<i>Continent</i>		
University population	91	49.2	Europe	78	42.2
General	63	34.1	North America	56	30.3
Farmer	13	7.0	Asia	25	13.5
Mixed	5	2.7	Africa	8	4.3
Children	4	2.2	Oceania	7	3.8
Elderly	3	1.6	South America	6	3.2
Capuchin monkey	1	0.5	Multiple	3	1.6
Unknown	5	2.7			

Notes: In two studies, the geographic location of the data is unknown. These studies were run online through Amazon's Mechanical Turk or a mobile app, and the authors did not specify what geographic controls were used.

and operations research (see the list of journals and their classifications in table A.2 in the online appendix).

We also identified where the data (either experimental or survey) were collected for 147 articles in the dataset. Most of these articles report estimates from data collected in a single country. Ten of them collected data from two to three countries/regions, and three of them (l'Haridon and Vieider 2019; Rieger, Wang, and Hens 2017; Wang, Rieger, and Hens 2017) conducted large-scale cross-country studies, collecting data from more than 30 countries/regions. In total, the estimates of the loss aversion coefficient comprised in our dataset come from 71 countries/regions (see figure D.1 in the online appendix).

Next we look at the basic design characteristics of the studies. We have 185 “studies”

reported in 150 papers, where a study is defined by a combination of several variables: type of the data, location of data collection, subject pool, type of reward, and continent of data collection. The frequency of each design characteristic is shown in table 1.

The majority of our data comes from laboratory experiments, but we also have studies using nonexperimental data such as surveys, stock market data, and game shows. Subjects were mostly recruited from the pool of university students or the general population. There is also a small set of studies that recruited special populations such as financial professionals, entrepreneurs, managers, and patients with psychiatric disorders or gambling problems. The type of reward used in the studies is mostly monetary. About three-quarters of the studies were conducted in Europe or North America.

TABLE 2
TYPES OF ESTIMATES

	All estimates		With SE	
	Frequency	Share (%)	Frequency	Share (%)
Aggregate-level	281	46.3	220	53.0
Individual-level mean	160	26.4	126	30.4
Individual-level median	166	27.3	69	16.6
Total	607	100.0	415	100.0

Notes: There are 85 cases where both mean and median of the distribution of individual-level estimates are reported. “With SE” indicates the observations where SEs are available. In addition, there are four aggregate-level estimates for which SEs are approximated with reported p -values, and 64 individual-level medians for which SEs are approximated with IQR. SEs are imputed for the rest of 124 observations (see section 4.2).

Next, we look at the main variable of interest, the estimated coefficient of loss aversion. We have a total of 607 estimates in the dataset (table 2). About half of these estimate the degree of loss aversion of a “representative” subject by pooling data from all subjects together (we call these aggregate-level, or simply aggregate, estimates). The other half estimated the coefficient for each individual subject in the study and reported summary statistics of the distribution, either mean or median. There are 85 cases where we have both the mean and the median of the distribution of the loss aversion coefficients estimated at the individual level.

Finally, we look at the specification of the functional form of U and the definition of loss aversion λ (table 3). There are 302 observations that assume the CRRA form for the utility functions as in equation (3), following Tversky and Kahneman (1992), but 221 of them assume and estimate a common curvature for gains and losses ($\alpha = \beta$). We observe less variation in the specification of reference points and the definition of loss aversion coefficients. Three-quarters

of the observations set the reference point at zero, but our dataset also includes studies where reference points are assumed to be subjects’ status quo or expectations. More than 80 percent of the observations estimate the loss aversion coefficient λ as Tversky and Kahneman (1992) define it.

4. Results

We structure the results into three distinct parts. We start with a nonparametric analysis of the reported loss aversion coefficients and their SEs. We subsequently fit random-effects meta-analytic distributions to the data and document the estimated mean loss aversion. Finally, we conduct a series of meta-regressions to see to what extent we can explain the estimated between-study variance.

4.1 Nonparametric Analysis

We start our presentation of the results by showing some nonparametric patterns in the reported loss aversion coefficient λ . Since we do not need SEs for this analysis, we can make use of all the estimates of loss

TABLE 3
UTILITY FUNCTION, LOSS AVERSION, AND REFERENCE POINT

	Frequency	Share (%)		Frequency	Share (%)
<i>Total number of estimates</i>	522	100.0			
<i>Loss aversion λ</i>			<i>Functional form of U</i>		
Tversky–Kahneman	445	85.2	CRRA	302	57.9
Köbberling–Wakker	37	7.1	CARA	53	10.2
Kőszegi–Rabin	9	1.7	Linear	73	14.0
Other	3	0.6	Other parametric	32	6.1
Not reported	28	5.4	Nonparametric	16	3.1
			Not reported	46	8.8
<i>Reference point</i>					
Zero	394	75.5			
Status quo	60	11.5			
Expectation	18	3.4			
Other/not reported	50	9.6			

Notes: There are 85 cases where both mean and median of the distribution of individual-level estimates are reported. We keep only one measure from each of these observations. See online appendix B for definitions of loss aversion.

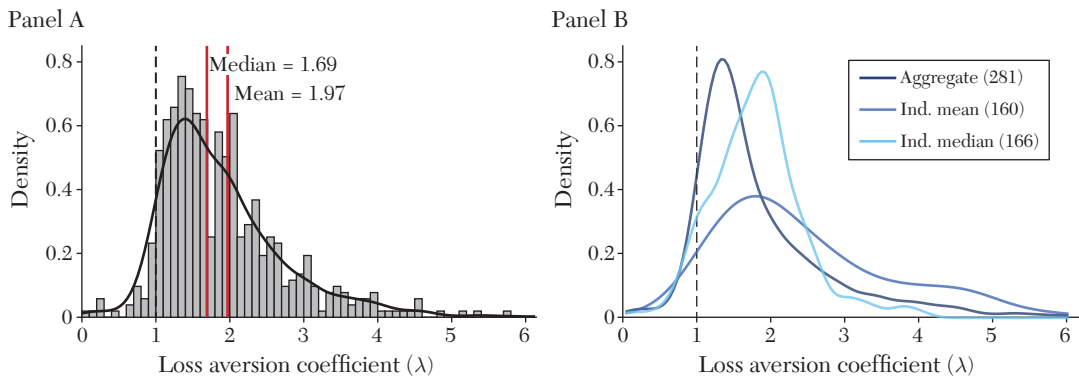


Figure 3. Distribution of Loss Aversion Parameters

Notes: (A) All reported estimates combined. (B) Separated by the type of estimates. Notes: panel A is identical to figure 2. In panel B, the kernel density estimate of the distribution of reported λ is plotted using the Gaussian kernel with Silverman's rule of thumb for the bandwidth selection. All 607 estimates in the data are used for estimation. In the online appendix, figure D.3 shows density plots of $\log(\lambda)$, figure D.4 shows histograms of λ separated by the type of estimates, and figure D.5 shows the empirical CDF of λ for each type of estimate.

aversion we coded in the dataset. Figure 3A shows the distribution of all the coded loss aversion parameters. The mean of the

parameters is 1.97. Given the right skew in the distribution, the median is considerably lower than the mean, at 1.69.

TABLE 4
SUMMARY STATISTICS OF REPORTED λ

Type	n	Mean	SD	Q1	Median	Q3	Min	Max
Aggregate-level	281	1.950	1.681	1.274	1.520	2.180	0.040	23.460
Individual-level mean	160	2.935	2.605	1.618	2.180	3.395	0.110	19.861
Individual-level median	166	1.844	0.756	1.417	1.800	2.090	0.110	7.500
All	522	1.970	1.370	1.310	1.690	2.288	0.040	23.460

Notes: There are 85 cases that report both individual-level mean and median. We keep individual-level medians from these cases in the last row of the table.

Figure 3B shows the same set of estimates, but now plots separate density functions for the estimates obtained from aggregate-level means ($n = 281$), individual-level means ($n = 160$), and medians ($n = 166$). Aggregate-level estimates can be seen to have the lowest mode (around 1.33), with individual-level medians having a slightly higher mode around 1.88. Means of individual-level estimates show a fat right tail, indicating a higher frequency of larger values. Table 4 shows summary statistics of reported λ for each type of measurement. The means of aggregate-level estimates and individual-level medians are close together, at 1.95 and 1.84, respectively. The somewhat lower mean of the latter results from fewer very large observations among medians than among aggregate estimates. The individual-level means have the largest variation ($SD = 2.61$ versus 1.68 for aggregate means and 0.76 for individual-level medians), including some of the smallest estimates as well as some of the largest. The truncated nature of the distribution then results in the highest mean by far, 2.94.

The differences between measurement types above cannot be interpreted causally. That is, the different measurements generally derive from different studies and are based on different data, so that the observed differences cannot be directly ascribed to the type of measurement used. To gain insight

into the effect of measurement type, we can conduct an analysis based on the 85 studies for which both means and medians are reported. With a mean of the means of 3.47 (median of means, 2.08) and a mean of the medians of 1.71 (median of medians, 1.69), the results confirm the ones for the overall sample (see figure D.6 in the online appendix). That is, the individual-level estimates tend to be rightward skewed, and this strongly affects the aggregate estimate reported in a paper when means of the individual-level estimates are used instead of medians. This issue, however, will be at least partly remedied by the observation that means of individual-level estimates also tend to come with increased SEs, which will, in turn, lead to increased pooling of the larger estimates in our meta-analytic estimations.

The earliest evidence recorded in our dataset is Tversky and Kahneman's (1992) famous 2.25. See figure D.2 in the online appendix for the time trend of reported estimates. Half of the estimates in the data are found in papers published after 2015. Individual-level estimates appear in the dataset after 2006, in part due to the rise of common experimental elicitation procedures. The raw data do not reveal a clear time trend, suggesting that estimates have remained the same on average for the last 30 years.

Just looking at the raw reported λ , figure 3 and table 4 suggest that the “average” loss

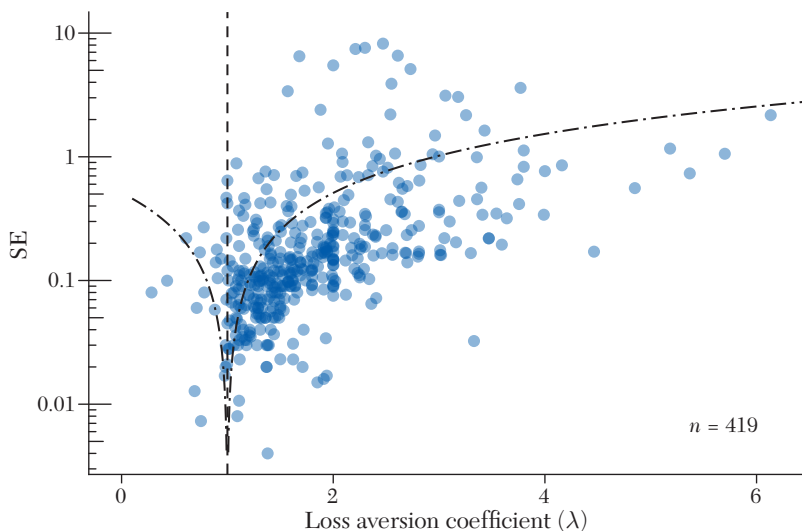


Figure 4. Relationship between Reported λ and Associated SE

Notes: Estimates with corresponding SEs reported are included ($n = 419$). For observations that have both individual-level mean and median, we keep the median in this figure. The vertical line corresponds to loss neutrality $\lambda = 1$. Two curves represent the boundaries for statistically-significant loss aversion ($\lambda > 1$) and loss tolerance ($\lambda < 1$). The x -axis is cut off at 6 and the y -axis is displayed in the log-scale for better visualization.

aversion coefficient λ would locate somewhere between 1.8 and 2.9. At the same time, there is high dispersion in the reported λ . These rough initial estimates, however, do not take the quality of the estimate into account. The latter can be assessed by means of the standard error associated with each estimate, which we can use to calculate a proper “meta-analytic average” by weighing each estimate by its precision, that is, the inverse of its standard error. Estimates falling far from the mean will then be given less weight to the extent that they have large associated standard errors.

4.2 Precision of Estimates

A common tool of meta-analysis is the so-called “funnel plot,” which visualizes the relationship between reported

estimates and their associated SEs (Stanley and Doucouliagos 2012). Figure 4 illustrates this relationship using data points where we have both a value for λ and an associated SE (either reported in the paper or reconstructed from other available information by us).

Figure 4 tells us several things about empirical estimates of loss aversion. Loss aversion may be imprecisely estimated, for instance, because of a small sample size, or because of other characteristics of the design. In our data, estimates that are far from 1 have higher SEs. They are not precisely estimated coefficients due to, for example, small sample size or other issues with the design or behavior. More precisely estimated λ 's tend to cluster between 1 and 2. Second, 398 out of 419 estimates

(95 percent) are larger than 1, producing a massive asymmetry in the funnel plot. Third, about 76.6 percent (305 out of 398) of $\lambda \geq 1$ estimates report results that are significantly different from 1 (based on the two-sided Wald test with the usual significance threshold of $p < 0.05$).

Table 2 shows that 192 out of our 607 recorded estimates are missing SEs. We approximate SEs from IQRs and p -values for 68 of these observations (see footnote 8), which leaves 124 SEs missing. Since standard errors are a fundamental ingredient for meta-analysis because they provide weights for the observations, we thus risk losing many observations, including the iconic measure of 2.25 reported by Tversky and Kahneman (1992). If studies not reporting any SEs are different from studies reporting them, we may furthermore distort our estimates systematically.

To overcome this issue, we impute the missing SEs using the subset of the data for which we have both λ and its associated SE. The basic idea is to estimate the parameters characterizing the distribution of SEs in the data, assuming that $\log(se)$ is drawn from a normal distribution $\mathcal{N}(\mu_{se}, \sigma_{se}^2)$ and that the mean depends on the size of reported λ , $\mu_{se} = \alpha_{se} + \beta_{se}\lambda$. We then impute missing SEs using the estimated distributional parameters $(\hat{\alpha}_{se}, \hat{\beta}_{se}, \hat{\sigma}_{se})$ and reported λ . See online appendix A.3 for the details. Online appendix C.3.2 also shows that our estimates and conclusions of this paper would not particularly change if these imputed values were dropped. We will, from now on, make use of the full set of observations.

4.3 Average Loss Aversion in the Literature

The main goal of our meta-analysis is first to obtain the “best available” estimate of the loss aversion coefficient λ combining the available information in the literature and then to understand the

heterogeneity of reported estimates across studies. Both goals can be informed by the data using a *Bayesian hierarchical modeling* approach.

4.3.1 Setup

Consider the dataset $(\lambda_i, se_i)_{i=1}^m$, where λ_i is the i th *measurement* (or *observation*) of the loss aversion coefficient in the dataset and se_i is the associated standard error that captures the uncertainty surrounding the estimate. In the benchmark model, we assume that the i th reported estimate λ_i is normally distributed around the parameter $\bar{\lambda}_i$:

$$(4) \quad \lambda_i | \bar{\lambda}_i, se_i \sim \mathcal{N}(\bar{\lambda}_i, se_i^2),$$

where the variability is due to the sampling variation captured by the known standard error se_i . The parameter $\bar{\lambda}_i$ is often referred to as the “true effect size” in meta-analysis.

Sampling variation is part of the observed variation in the reported estimates $(\lambda_i)_{i=1}^m$, but in addition, there may be “genuine” heterogeneity across measurements (due to different settings, for example). We model this by assuming that each $\bar{\lambda}_i$ is, in turn, normally distributed, adding another level to the hierarchy:

$$(5) \quad \bar{\lambda}_i | \lambda_0, \tau \sim \mathcal{N}(\lambda_0, \tau^2),$$

where λ_0 is the *overall mean* of the estimated loss aversion parameters $\bar{\lambda}_i$, and τ is its standard deviation, capturing the variation between observations in the data. The overall variance in the data, therefore, consists of two parts, the between-observation variance, τ^2 , and the individual sampling variance coming from measurement uncertainty, se . This can be clearly seen by combining expressions (4) and (5) into one:

$$\lambda_i | \lambda_0, \tau, se_i \sim \mathcal{N}(\lambda_0, \tau^2 + se_i^2).$$

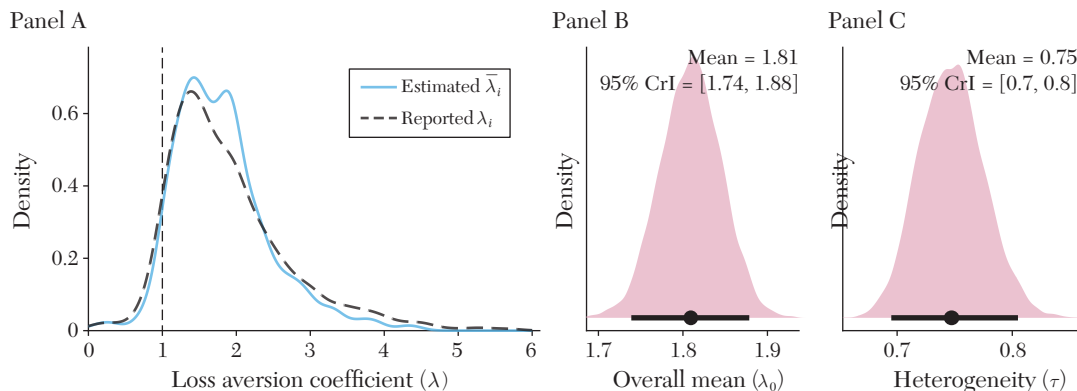


Figure 5. Estimation Result of Model M1

Notes: (A) Density plot of loss aversion coefficient, estimated $\bar{\lambda}_i$ versus reported λ_i . (B) Posterior draws of the overall mean λ_0 . (C) Posterior draws of the heterogeneity parameter τ . The dashed curve in panel A indicates the density of the observed loss aversion parameters, λ_i ; the solid curve in panel A indicates the density of the estimated parameters, $\bar{\lambda}_i$. Observations above six are not shown in panel A for better visualization. The black dots and lines in panels B and C represent the posterior means and the 95 percent credible intervals of λ_0 and τ , respectively.

4.3.2 Model Estimation

We start from fitting the model expressed as equations (4) and (5), restated as model M1 here, to the data $(\lambda_i, se_i)_{i=1}^m$:

$$\begin{aligned}
 \text{(M1)} \quad \lambda_i | \bar{\lambda}_i, se_i &\sim \mathcal{N}(\bar{\lambda}_i, se_i^2), \\
 \bar{\lambda}_i | \lambda_0, \tau &\sim \mathcal{N}(\lambda_0, \tau^2), \\
 \lambda_0 &\sim \text{half } \mathcal{N}(1, 5), \\
 \tau &\sim \text{half } \mathcal{N}(0, 5),
 \end{aligned}$$

where “half $\mathcal{N}$ ” indicates the half-normal distribution that “folds” the normal distribution $\mathcal{N}(0, \sigma^2)$ at its mean to have nonzero probability density for values greater than or equal to 0. This model incorporates the assumption that every observation is statistically independent, and that the observations are normally distributed. We will relax these rather strong assumptions in due time.

We estimate the model in Stan (Carpenter et al. 2017) using Hamiltonian Monte Carlo simulations, and launch it from R (R Core Team 2020) using RStan (Stan Development Team 2020). We chose half-normal distributions with a standard deviation of 5 for priors for the population-level parameters λ_0 and τ , but the estimates are not sensitive to changing the prior, given the amount of data we have (see online appendix C.3.3).

The estimated overall mean λ_0 is 1.809 with a 95 percent credible interval (CrI) of [1.739, 1.878].^{10,11} The mean is clearly lower than the non-parametric result that

¹⁰A Bayesian credible interval (CrI) of size $1 - \alpha$ given data D is an interval $[L(D), U(D)]$ such that $\Pr(L(D) \leq \theta \leq U(D)) = 1 - \alpha$, where θ is the parameter of interest. Unlike the (frequentist) confidence interval, CrI has a literal probabilistic interpretation: given the data, there is a $100 \times (1 - \alpha)$ percent probability that the true parameter value is in the interval.

¹¹Results from the frequentist random-effects meta-analysis are presented in online appendix E. We obtain largely identical estimates.

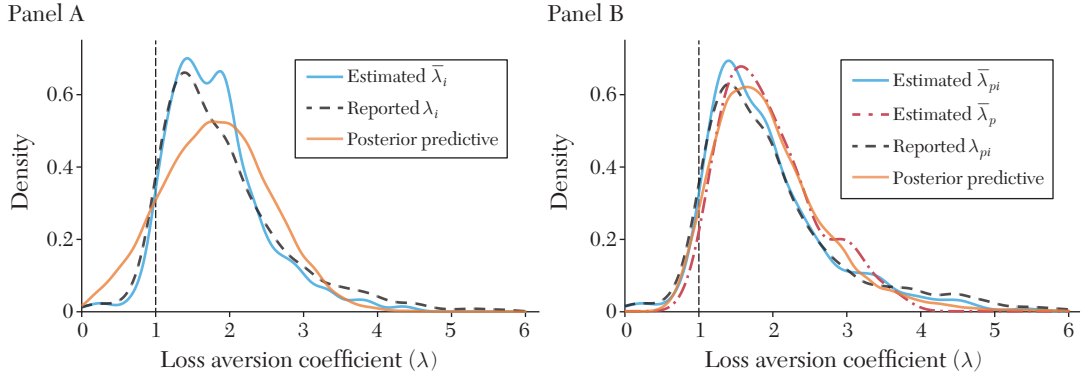


Figure 6. Distributions of Reported and Estimated λ , and Posterior Predictive Distribution of λ .

Notes: (A) Assuming a normal distribution for the population level (model M1). (B) Assuming a log-normal distribution for the population level (model M2).

we saw above, which was 1.97. This is shown in figure 5. The density of estimated $\bar{\lambda}_i$ is lower than the one of observed λ_i for values above 2.5. The same occurs for values below one. This is meta-analytic pooling at work—estimates that fall far from the mean are shrunk toward more plausible values, with the amount of shrinkage proportional to the standard error. See discussion in online appendix C.2.

The estimates produced are, of course, only valid conditional on our assumptions. We already know that the normality assumption seems a stretch, given the skewed distribution of the reported λ . To see this, we can take a look at the *posterior predictive distribution*—the distribution of loss aversion coefficients we would expect new observations λ_{new} to display, provided that the characteristics of the studies from which these observations are obtained are similar on average to those of past studies—and compare it to the distribution of actual observations.¹² This is shown in figure 6, panel A. Relative to the

actual observations—either as reported (λ_i), or as estimated ($\bar{\lambda}_i$)—the posterior predictive distribution overestimates the likelihood of values smaller than 1, while it underestimates the likelihood of intermediate values between 1 and 2. It does not attribute any probability to values beyond 4, which are not uncommon in the data. It thus seems desirable to look for a model that may provide a better fit to the data.

We thus extend the baseline model M1 in two ways. First, we use a log-normal distribution for the population-level distribution. Second, we explicitly model the nesting of observations in papers, in order to overcome any potential distortions deriving from nonindependence of observations. Remember that our 607 observations have been obtained from 150 distinct papers, the largest number

$$\pi(\lambda_{new} | (\lambda_i)_{i=1}^m) = \int \pi(\lambda_{new} | \theta) \pi(\theta | (\lambda_i)_{i=1}^m) d\theta,$$

where $\theta = ((\bar{\lambda}_i)_{i=1}^m, \lambda_0, \tau)$ is a vector of model parameters (Gelman et al. 2014). Evaluating this integral is difficult, but we approximate it by drawing $\lambda_{new}^{(s)} \sim \mathcal{N}(\lambda_0^{(s)}, \tau_{(s)}^2)$ using posterior simulations $(\lambda_0^{(s)}, \tau_{(s)})$, $s = 1, \dots, N$. We have 8,000 draws (2,000 iterations $\times$ 4 chains) of λ_{new} .

¹²Formally, the posterior predictive distribution is written:

of observations in a single paper being 53 (Rieger, Wang, and Hens 2017; Wang, Rieger, and Hens 2017). The independence assumption seems rather heroic in this case. To do this, we introduce paper-level estimates as an additional hierarchical level. Let λ_{pi} be the i th estimate reported in paper p . We formulate a model as follows:

$$\begin{aligned}
 \text{(M2)} \quad & \lambda_{pi} | \bar{\lambda}_{pi}, se_{pi} \sim \mathcal{N}(\bar{\lambda}_{pi}, se_{pi}^2), \\
 & \bar{\lambda}_{pi} | df, \bar{\lambda}_p, \sigma_p \sim t(df, \bar{\lambda}_p, \sigma_p^2), \\
 & \bar{\lambda}_p | \lambda_0^\ell, \tau_\ell \sim \log \mathcal{N}(\lambda_0^\ell, \tau_\ell^2), \\
 & \lambda_0^\ell \sim \mathcal{N}(1, 5), \\
 & \tau_\ell \sim \text{half } \mathcal{N}(0, 5), \\
 & df \sim \text{half } \mathcal{N}(0, 5), \\
 & \sigma_p \sim \text{half } \mathcal{N}(0, 5).
 \end{aligned}$$

The model M2 now explicitly nests the estimated observation-level parameters, $\bar{\lambda}_{pi}$, in paper-level estimates, $\bar{\lambda}_p$. The former are modeled as following a robust student- t distribution instead of a normal distribution to account for observed outliers.¹³ The latter are modeled as following a log-normal distribution. Note the super- and subscripts ℓ in the location and scale parameters $(\lambda_0^\ell, \tau_\ell^2)$ of the log-normal distribution. We can calculate the mean and the median of the distribution by $\exp(\lambda_0^\ell + \tau_\ell^2/2)$ and $\exp(\lambda_0^\ell)$, respectively, exploiting the properties of the log-normal distribution.

We again start by examining the fit of the model to the data, and by summarizing the

population-level parameters. The model fit is shown in figure 6B. The log-normal distribution can now be seen to fit the estimated paper-level data well. The distribution of the paper-level observations has more probability mass between about 1 and 3, but less beyond that point, compared to the actual study-level observations. The degrees of freedom of the student- t distribution are estimated at 1.32, thus vindicating the use of the robust distribution. The mean loss aversion parameter obtained from this estimation is 1.955, with a 95 percent CrI of [1.820, 2.102]. Notice, however, that even though this estimate is nearly identical to the one obtained under the standard model at the outset, that occurs by coincidence rather than being a feature of the model. One can further see that there is now increased uncertainty surrounding the prediction interval. This is indeed natural, since the paper-level estimates are surrounded themselves by larger amounts of uncertainty, which is then passed up the hierarchy to the aggregate parameters.

4.3.3 Robustness Checks

Online appendix C.2 presents estimation results for two additional models, but our preferred model M2 fits better than these “intermediate” models. We also estimated the models under different priors or using the “complete” data including only observations where associated SEs are available and obtained similar conclusions. These robustness checks are presented in online appendix C.3.

4.3.4 Heterogeneity between Studies versus between Individuals

An interesting question concerns how the heterogeneity between studies we document compares to typical levels of heterogeneity between individuals. While our data are ill-suited to answer this question in general, figure 7 provides an indication by comparing the study-level estimates of

¹³We estimate the degrees of freedom of the distribution, df , endogenously from the data. This allows us to determine whether the student- t distribution provides a good fit, which is the case if the degrees of freedom are small, or whether it converges to a normal distribution, which is the case for large degrees of freedom (Kruschke 2010).

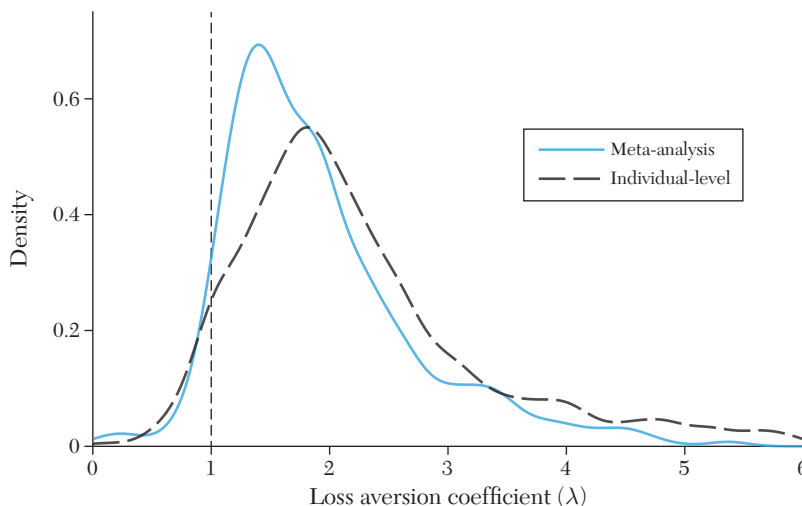


Figure 7. Distributions of Estimated $\bar{\lambda}_{pi}$ Compared to Individual-Level Estimates from l’Haridon and Vieider (2019)

Notes: Estimates from l’Haridon and Vieider (2019) follow the original hierarchical setup with country-level fixed effects reported in that paper.

loss aversion obtained from model M2, $\bar{\lambda}_{pi}$, to the between-subject distribution of loss aversion in the data of l’Haridon and Vieider (2019), containing estimates for 3,000 students from 30 countries. Both distributions display a log-normal shape, and have peaks in their densities between 1 and 2. The individual-level estimates are somewhat more dispersed, having slightly more probability mass on small values below 1, as well as more probability mass on values above 2.5. Overall, however, the two distributions are similar. This illustrates just how much heterogeneity we find between studies, which may arguably be driven at least in part by differences in experimental designs, model definitions, and estimation methods.¹⁴

¹⁴We note one finding that shows strong between-subject heterogeneity and its association with IQ. Chapman et al. (2018) report data from an incentivized representative survey of Americans measuring loss

The comparison shown here comes with a large caveat—individual-level distributions from other papers may look very different. This also goes back to the point on study quality we made above. Given suitable restrictions on choice lists and modeling assumptions, it would be easy to produce individual-level distributions that are narrower than the one shown here. On the other hand, wide choice lists, general definitions,

aversion and other behavioral parameters. In the representative survey, the median λ is 0.99. In a college sample using a highly similar protocol, the median λ is 1.84. In the survey of Americans there are substantial correlations (around 0.2-0.3) between IQ and loss aversion. This paper is not included in our meta-analysis because it fell outside our time window. However, it uses both a different design (optimized adaptive estimation) and features several important within-study measures of heterogeneity, especially IQ. Since this is the only study to report cross-IQ heterogeneity, meta-regression of that feature will add little to our general conclusions. Such a study could therefore feature prominently in a narrative review (for reasons discussed in our conclusion).

TABLE 5
SUMMARY OF ESTIMATION RESULTS

Model	Distributional assumption			Posterior of λ_0				Posterior of τ			
	Obs. level	Paper level	Pop. level	Mean	SD	2.5%	97.5%	Mean	SD	2.5%	97.5%
M1	Normal		Normal	1.809	0.036	1.739	1.878	0.746	0.028	0.695	0.803
M2	Normal	Student- <i>t</i>	Log-normal	1.955	0.072	1.820	2.102	0.743	0.342	0.604	0.904

Notes: In Model M2, (λ_0, τ) are calculated from the log-normal parameters $(\lambda_0^\ell, \tau_\ell)$ by $\lambda_0 = \exp(\lambda_0^\ell + \tau_\ell^2/2)$ and $\tau^2 = [\exp(\tau_\ell^2) - 1]\exp(2\lambda_0^\ell + \tau_\ell^2)$.

and noisiness in measurements can all contribute to much wider distributions. It can indeed be shown that, given a wide enough range of possible estimates, the location parameter of the log-normal posterior predictive distribution fit to the individual-level estimates will decrease systematically as the proportion of random choices increases, while the dispersion will increase in random choices.¹⁵ This goes to show just how many different factors may impact the estimation of loss aversion coefficients.

4.4 Explaining Heterogeneity

We observe a non-negligible amount of between-paper heterogeneity (expressed in estimated τ in table 5, model M2) among reported estimates of λ . In this section, we seek to understand the source of this variability in order to provide a tentative answer to our second key question: “Do reported estimates of λ systematically vary by underlying design characteristics for measurement of loss aversion?”

Remember that we coded several features about the characteristics of the study design (table A.1 in the online appendix). Figures D.10 and D.11 in the online appendix provide a first look into how these design features are related to reported estimates of λ . Each panel presents how the reported λ varies by underlying design characteristics. We do observe some patterns in the figure, but the effects appear rather weak and it is not clear if these relations are systematic and robust.

We approach this question with a *random-effects meta-regression*, which extends our previous random-effects model by incorporating coded features of the observation or the paper into the model. More precisely, we set up a new model, which expands model M2 by allowing for the location of the observations to be systematically shifted depending on observed characteristics of the observation or the paper. The model looks as follows:

$$\lambda_{pi} | \bar{\lambda}_{pi}, se_{pi} \sim \mathcal{N}(\bar{\lambda}_{pi}, se_{pi}^2),$$

$$\bar{\lambda}_{pi} | df, \bar{\lambda}_p, \sigma_p, \beta \sim t(df, \bar{\lambda}_p + \mathbf{X}_{pi}\beta, \sigma_p^2),$$

$$\bar{\lambda}_p | \lambda_0^\ell, \tau_\ell \sim \log \mathcal{N}(\lambda_0^\ell, \tau_\ell^2),$$

$$\lambda_0^\ell \sim \mathcal{N}(1, 5),$$

$$\tau_\ell \sim \text{half } \mathcal{N}(0, 5),$$

¹⁵Note that the range one can theoretically estimate will depend both on the measurement obtained and the definition of loss aversion. For instance, a narrow choice list in the mixed domain ($x > 0 > y$) may come up with a narrow range of estimations if the definition of loss aversion is taken to be $\lambda = x/(-y)$. For a definition $\lambda = U(x)/[-U(y)]$, however, the range resulting from that same choice list could be very wide if the measurements of U allow for extreme values.

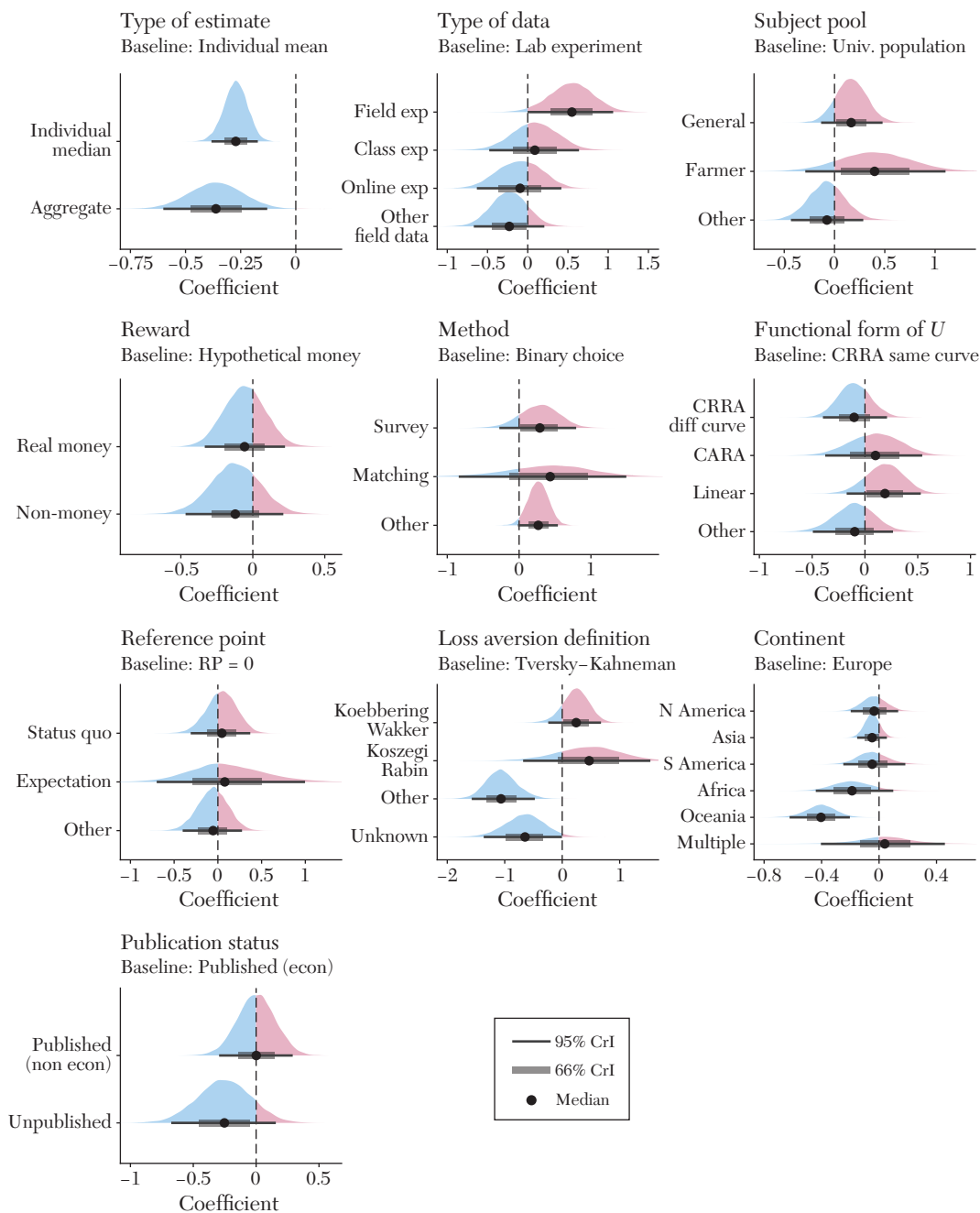


Figure 8. Bayesian Random-Effects Meta-regression

Notes: Posterior distributions of coefficients β , together with posterior medians (black dot), 66 percent (thick solid line) and 95 percent (thin solid line) credible intervals, are shown. Table D.1 in the online appendix presents the result in a table format.

$$df \sim \text{half } \mathcal{N}(0, 5),$$

$$\sigma_p \sim \text{half } \mathcal{N}(0, 5),$$

where $\mathbf{X}_{pi}$ is a vector of study characteristics associated with i th observation reported in paper p . These characteristics consist mostly of dummy variables taking the value of 0 or 1, with β a vector of coefficients. To facilitate the interpretation of the constant, non-dummy independent variables included in vector $\mathbf{X}_{pi}$ are mean-centered—each coefficient in the vector β then captures the additive effect on the paper-level mean $\bar{\lambda}_p$, relative to the “baseline study” (characterized by the omitted categories in dummy variables and the means of non-dummy variables), which will become clear later.

Estimation results are presented in figure 8. First, the posterior mean of the estimated $\bar{\lambda}_{p^*}$ for the benchmark study p^* is 1.981 (95 percent CrI = [1.582, 2.380]). Each estimated coefficient in β captures the effect of the study characteristic from this benchmark value.

As we have seen in section 4.1, the type of estimates reliably captures the variation in reported λ —individual-level means tend to be higher than the other two types of estimates due to skewed distributions of individually estimated λ . We also find that field experiments are associated with higher λ compared to laboratory experiments and studies recruiting general population samples are also associated with higher values of λ compared to the studies with a population of university students. We do not observe differences between studies using monetary rewards and nonmonetary rewards, but survey studies tend to produce lower estimates of λ than the binary lottery choice tasks that are common in laboratory experiments. In terms of the specification of the value function, it does not seem to matter much which functional form (CRRA, CARA, etc.) one assumes for the utility function U ,

or whether reference points are assumed to be zero, status quo, or expectations. Studies estimating λ following the definition by Köbberling and Wakker (2005) produce higher λ compared to the standard Tversky and Kahneman’s (1992) definition, but the effect is modest.¹⁶

Taken together, our Bayesian meta-regression analysis uncovers some factors that are associated with the size of reported loss aversion coefficients, but it is still a difficult task to draw a complete picture of the observed heterogeneity. We note that 15.1 percent of the between-observation variance is explained by covariates.

4.5 Publication Bias

The cumulation of scientific knowledge is threatened by selective reporting or publication of findings. For example, suppose a theory or body of evidence makes a strong prediction about the sign or magnitude of a certain effect. Selective reporting occurs if scientists, editors, and reviewers believe effects to be the norm, and there is a bias against reporting or publishing “unusual” results that contradict the norm. Selective reporting of this kind slows down the crucial process of scientific self-correction.

We will refer to such selective reporting of scientific findings collectively as “publication bias.” Publication bias can take on many forms. We discuss two possibilities of relevance for the case of loss aversion. The first case is when journals prefer to see numerical estimates of loss aversion close to a certain number, and are skeptical of estimates that

¹⁶We have 10 estimates of λ reported in eight papers that use the definition of loss aversion according to Köszegi and Rabin (2006, hereafter KR). Since the KR formulation incorporates consumption utility (as we discuss in section 2), caution is needed when comparing estimates of λ using the KR formulation and other estimates of λ following the standard Tversky and Kahneman (1992) definition. DellaVigna (2018), for example, suggests that a loss aversion of 2.25 in Tversky and Kahneman (1992) translates into a loss aversion of 3.35 in KR (footnote 16, p. 674).

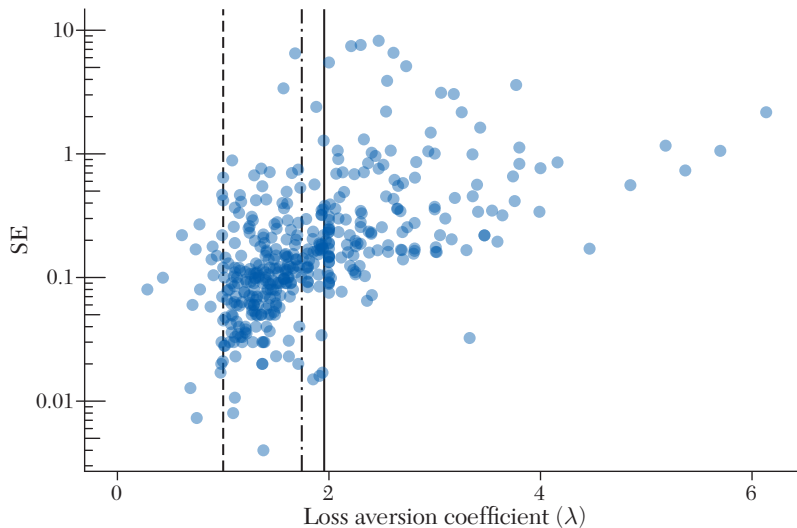


Figure 9. Funnel Plot

Notes: The vertical dashed line corresponds to no loss aversion $\lambda = 1$. The vertical solid line corresponds to the estimated mean λ_0 from model M2, 1.955. The vertical dash-dotted line corresponds to the bias-corrected mean λ from model M1-ext, 1.743. The x -axis is cut off at 6 and the y -axis is displayed in the logarithmic scale for better visualization.

deviate far from that number. For there to be a bias, the preferred estimate by journal editors must differ from the “true” estimate found in the population of studies (see Borenstein et al. 2009).

A second form of bias concerns the overall significance of results. In this form, journals prefer results that are “statistically significant” (usually demarcated by a p -value of 0.05) and reject some null hypothesis (Andrews and Kasy 2019; Brodeur et al. 2016; Brodeur, Cook, and Heyes 2020; Chopra et al. 2024). From the viewpoint of the authors of this paper, there is not a single specific test that can address these forms of publication bias jointly. Even if there were, the particular issues with this data might make it difficult to trust a single test absolutely. Instead we will take a descriptive approach to both types of publication bias.

Regarding the first form, in the context of loss aversion, one might suspect that researchers preferentially report evidence for loss aversion ($\lambda > 1$) and put evidence for loss tolerance ($\lambda < 1$) “in the file drawer” because such results contradict the initial hypothesis (Rosenthal 1979). Other sources of publication bias are possible. Researchers in some disciplines may be motivated to undermine the “prevailing paradigm” of loss aversion and preferentially create or publish low- λ results. Alternatively, some journals may be interested in publishing results that conform to the pre-prospect theory economic orthodoxy of no loss aversion. Even coming up with a null hypothesis is thus more complex in our case than it would be when trying to simply ascertain the effect of a treatment or its absence.

Funnel plots (see section 4.2) are often used as a tool to examine bias from the results of meta-analyses (Egger et al. 1997, Stanley and Doucouliagos 2010). Figure 9 shows such a plot for all data points for which we have both an estimate of loss aversion and an associated standard error (thus excluding estimates with imputed SEs). In the absence of publication bias, the observations at the bottom of the graph, which have higher precision, should be concentrated around the underlying mean estimate, indicated by the solid vertical line. As we move up in the graph toward the top of the funnel and the precision of the studies decreases, we expect an increase in the degree of dispersion around the mean estimate. In the absence of publication bias, this dispersion ought to be symmetric around the mean. A larger number of observations in the upper right side of the graph compared to the upper left side would then be an indicator of classical publication bias, whereby estimates of loss aversion that fall closer to 1 and are not significant are less likely to be reported.

At first sight, there would indeed appear to be such an asymmetry in the graph. We clearly observe some large estimates in the upper right part of the graph, and hardly any corresponding estimates in the upper left part. Even more pronounced is the large cluster of studies at the bottom left corner of the graph. One might be concerned they indicate a model of publication bias where the true λ is around 0.8–1, but editorial bias favors publication of studies with higher estimates at lower precision. (In such a model, low-precision, low-estimate studies would not be published because of the judgment of journal referees or editors, and high-precision, high-estimate studies would not be published because they are statistically improbable.)

To test asymmetry in the funnel plot through examining the correlation between reported estimates and their SEs, we extend

the benchmark model M1 by allowing the underlying mean for each reported estimate to depend systematically on the standard error:

$$\begin{aligned} \text{(M1-ext)} \quad \lambda_i | \lambda_0, \gamma, \tau, se_i &\sim \mathcal{N}(\lambda_0 + \gamma \\ &\times \sqrt{\tau^2 + se_i^2}, \\ &\tau^2 + se_i^2), \end{aligned}$$

where γ represents a potential publication bias and $\lambda_0 + \gamma\tau$ captures the average loss aversion coefficient correcting for the bias (since $E[\lambda_i] \rightarrow \lambda_0 + \gamma\tau$ as precision goes to infinity, or equivalently, $se_i \rightarrow 0$).¹⁷ We find that the estimated mean γ is positive (i.e., 1.427 with a 95 percent CrI of [0.986, 1.872]), consistent with publication bias of a specific form: higher λ are less precisely estimated and “small” λ are hidden in the literature. The “corrected” mean λ is the mean $\lambda_0 + \gamma\tau$ which is 1.743 with a 95 percent CrI of [1.672, 1.812]. Thus, the model suggests a relatively milder publication bias than figure 9 might suggest, but nonetheless a drop from the average λ of our main estimate of 1.955 (model M2, table 5).

The use and appropriate interpretation of funnel plot asymmetry has been debated, and there are reasons to not take this estimate at face value. A crucial assumption in measuring publication bias using these tests is that there is no correlation between estimates and their standard errors (in the absence of publication bias and between-study heterogeneity). In the context of the estimation of behavioral parameters in experimental

¹⁷This model is motivated by the regression-based approach for detecting and correcting for publication bias in meta-analysis, introduced first by Egger et al. (1997) and established by Card and Krueger (1995), Stanley (2005, 2008), and Stanley and Doucouliagos (2014). Here we follow the “extended random-effects model” developed by Rucker et al. (2011), to be consistent with the hierarchical model we set up in section 4.3.

economics like ours, there are plausible reasons why the no-correlation assumption might fail. First, it is possible that researchers choose parameters in their experiments (such as a series of monetary outcomes used in a multiple price list) in a way that is tuned to detect loss aversion coefficients that are close to 1 or 2. Second, the parameter λ must be larger than 0 by construction of the theory, and the reported estimates exhibit non-normality. In each of these cases, the correlation between estimates and standard errors can arise “mechanically” even without any publication bias.¹⁸

A different way to examine a certain type of publication bias is to compare values reported in published and unpublished papers. This method does not rely on assumed independence between estimates and their standard errors. Publication bias by journals implies that the estimates found in published papers could vary from estimates in unpublished working papers. Our meta-regressions (presented in section 4.4) estimated that observed loss aversion coefficients were 0.25 lower in working papers than in published studies, roughly 1.73 for working papers (though the two credible intervals are overlapping). Remember that the “true” value of λ is best estimated by a weighted average of the working paper estimate and the published paper estimate, so 1.73 can still be thought of as a lower bound of the “true” estimate. It is perhaps reassuring that this value is almost the same as the bias-corrected estimate of 1.74 discussed previously.

A second form of publication bias involves a journal focusing too much on “statistically

significant” results. In this form, journals prefer to publish results that reject the null hypothesis for the parameter of loss aversion, such as $\lambda = 1$ or $\lambda = 2$, which could result in a distribution of test statistics that exhibits a discontinuity around a threshold for statistical significance (such as $z = 1.96$). Gerber and Malhotra (2008a, b) introduced a “caliper test” to identify a systematic bunching in the distribution of test statistics within narrow bands around the threshold of statistical significance, but such an approach requires more observations than we have in our sample. Absent the ability to perform that test, we provide histograms of the z -statistics of our estimates in figure 10. There are two histograms, centered around the null hypotheses of $\lambda = 1$ (panel A) and $\lambda = 2$ (panel B). The figure does not appear to reveal any such spikes in the positive or negative direction around values of 1 or 2 (which would appear around the vertical dotted line representing zero deviation), regardless of which null hypothesis we consider.

5. Discussion

Loss aversion is an important concept in behavioral economics and has been applied widely. This paper reports a meta-analysis of empirical estimates of the loss aversion coefficient λ . Our preferred specification indicates a mean $\lambda = 1.955$ and a 95 percent credible interval of $[1.820, 2.102]$. Many other specifications are within 0.1–0.3 of this finding and produce credible intervals that do not include 1 or 2.25. The former number is consistent with no loss aversion; the latter is an early estimate from Tversky and Kahneman (1992) which seems a bit too high. While there is a wide degree of heterogeneity across estimates, in general, no single factor emerges from meta-regression that greatly changes estimated loss aversion. Estimates derived from nonuniversity

¹⁸A similar point was raised by Matoušek, Havránek and Iršová (2022), who did a meta-analysis of experimentally measured individual discount rates, which are typically bounded at zero. See also Sterne et al. (2011) for more general discussion and recommendations for the interpretation of funnel plot asymmetry in the context of meta-analyses in RCTs.

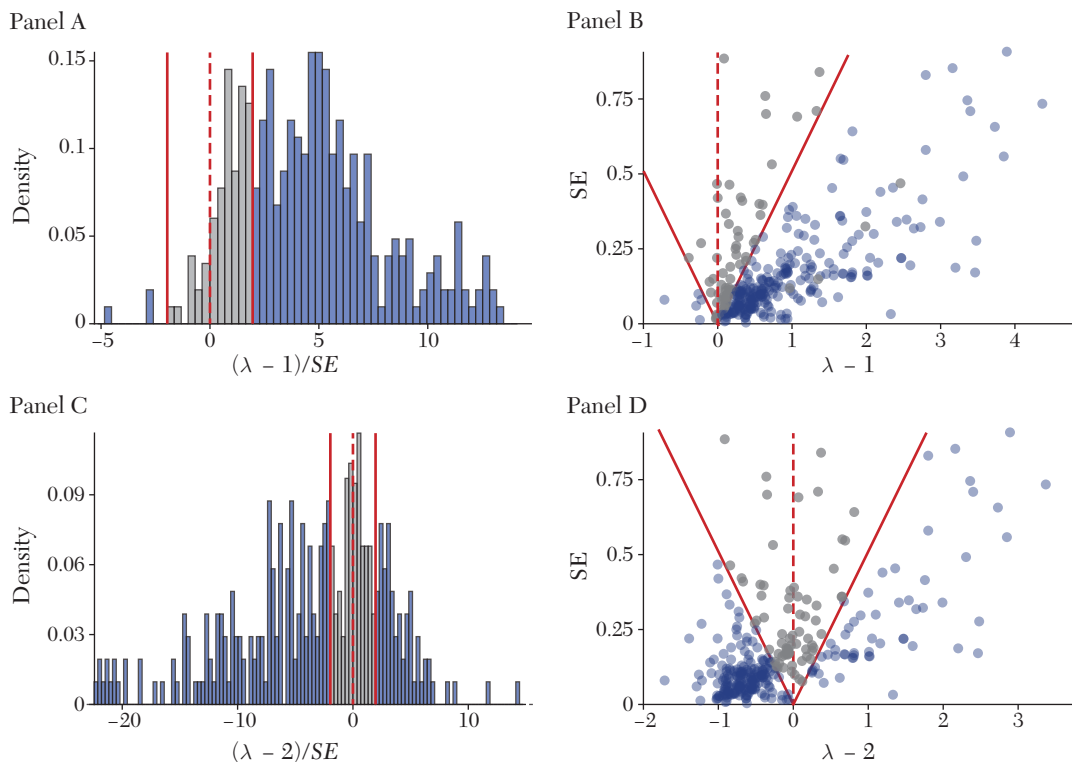


Figure 10. Examining Publication Bias

Notes: The left panels (A, C) show binned density plots for the z -statistics and the right panels (B, D) plot the estimated “effect size” against its standard error. In panels A and B, the effect size is $\lambda - 1$, corresponding to the null hypothesis of $H_0: \lambda = 1$. In panels C and D, the effect size is $\lambda - 2$, corresponding to the null hypothesis of $H_0: \lambda = 2$. *Notes:* The plots include 346 observations of aggregate-level and individual-level mean λ , which have associated standard errors reported in the paper. The solid red lines mark $|z| = 1.96$. Blue bars and dots correspond to effect sizes that are significantly different from zero at the 5 percent level. The bin width is 0.32 in panels A and C. Outliers are not shown in the plots: the x -axis is restricted to the interval $[Q1 - 1.5 \times \text{IQR}, Q3 + 1.5 \times \text{IQR}]$ in panels A and C and both axes are cut off at $Q3 + 3 \times \text{IQR}$ in panels B and D. Figure D.8 in the online appendix shows the “full” version of panels B and D, including outliers.

populations, field experiments, and means of individual elicitations (compared to aggregates) are correlated with a modest increase in the loss aversion parameter.

A main takeaway from this paper is that the point estimate of $\lambda = 1.955$ reported above—estimated jointly with the

uncertainty surrounding it—is the best current answer to the question of how large the loss aversion coefficient truly is. It is not the ultimate answer, however. One can indeed easily update this information with new evidence, either by using this as a Bayesian (hyper-)prior in subsequent estimations of

loss aversion, or by combining the posterior from our meta-analysis with equivalent evidence from a follow-up meta-analysis on studies not yet included into our sample, e.g., because they appeared after our cutoff date.

We note two possible limitations to our study and, wherever possible, note where successive studies could improve. First, a key concern with any empirical analysis is a differential selection of reported data. The outlined criteria for inclusion in our dataset are explicit and objective; we have no reason to believe, *ex ante*, that it should be correlated with our parameter of interest. However, our dataset is dominated by published studies. To the extent that publication may vary with a reported λ parameter, our analysis may suffer from bias.¹⁹ Because the parameter of interest here is bounded by zero and positively skewed, it is difficult to use standard meta-analysis techniques like funnel plot (see figure 9) and regressions to measure publication bias. Further, an important test of publication bias is whether unpublished working papers report reliably different results than published ones. There appears to be a modest difference; unpublished papers are about 0.25 lower which, notably, is still within the credible interval of our current estimate, and may be due to chance, given the small sample. While our preferred estimate is $\lambda = 1.955$, based on the assumption that available studies are unbiased, researchers more concerned about publication bias may want to consider lower values (e.g., $\lambda = 1.74$).

Second, while we followed conventional practices of coding study quality (see section 3.2), medical and health meta-analyses follow a more formal method to assess study quality as “risk of bias.” Tens of thousands

of Cochrane meta-analyses and systematic reviews follow a standard protocol where multiple raters evaluate risks of bias (study quality) on multiple dimensions such as the randomness of the treatment assignment and the blinding of participants and researchers, among others.²⁰ For instance, Hollands et al. (2015) is a meta-analysis of experiments testing the causal effect of changing portion, package, or tableware size on how much food, alcohol, or tobacco is chosen or actually consumed. Their figure 3 shows that of their 72 studies assessed, about one-third were low risk, the modal percentage were unclear risk, and less than 10 percent were deemed high risk.²¹

Meta-regression analysis is uniquely able to adjust for unobserved differences in research quality across studies. If every study in our dataset contained multiple estimates, a fixed-effect panel meta-regression would have eliminated any potential distortion that differences in study quality may have on research findings (Stanley and Doucouliagos 2012, pp. 112–17). Outside of this special case, the meta-regression can only control for effects and is dependent on how the values of quality are coded. Accordingly, we have recorded, for each estimate, the impact

²⁰ Risk of bias, introduced in 2008, is a tool widely used in Cochrane reviews and other systematic reviews for assessing risk of bias in randomized trials (Higgins et al. 2011). Their general criteria are: randomness, blinding of participants and personnel (to treatment), blinding of outcome assessment, incomplete outcomes (attrition), selective reporting, comparability of groups (a.k.a. balance check), and consistency of intervention delivery. In each of these categories there are a small number of precise questions that are subjectively rated by assessors on a four-point categorical scale (yes, probably yes, probably no, no) and then aggregated. See Sterne et al. (2019) for a revised tool (RoB 2).

²¹ A possible analogue exists within social science research (especially for experiments), forecasts of likely replicability of a study’s findings. Altmejd et al. (2019) use data from actual replications to fit a machine learning model. They are able to predict the degree of replicability with about 75 percent accuracy from simple observable features of a study.

¹⁹ It is important to note that any research synthesis technique would suffer from this issue, not just meta-analysis (Borenstein et al. 2009).

factor of the journal in which they were published (if applicable). The impact factor value has little explanatory power in regard to our results. We note (1) there is no meaningful correlation between journal impact factor and reported (mean) λ value; (2) inclusion or exclusion of journal impact factor does not meaningfully change our meta-regression results. To the extent that journal impact factor can be a proxy for study quality, we see no evidence to support the concern that “low” quality studies are biasing our results.

In the introduction we promised to return to subtle, important comparisons between meta-analysis and narrative review. We will do that now.

Many readers will not know much about meta-analysis and might be skeptical about it, especially in comparison to the familiar style we call “narrative review.” We anticipate such skepticism and offer counterarguments. The unabashed goal is to advocate for more appreciation of the underdog method of meta-analysis.

We will start with fears about narrative review. Such reviews could be influenced by biases in remembering recent (and perhaps socially connected) salient, recently encountered data. Meta-analysis is a partial antidote.

There are two natural fears about meta-analysis. One is that it is somehow a mistake to reduce all research in a field to a single value or a range of numbers. The second fear is that simple apples-to-apples comparisons across studies neglect heterogeneity of methods (see Borenstein et al. 2009, for a history of such critiques). These fears are natural, but meta-analysis techniques have evolved to allay such fears.

The “single number” fear is misguided because humans—including scientists—value simplicity. If meta-analysis did not provide a carefully and transparently derived value (or range), researchers will imagine another simplified value, one way

or another. It is also often necessary to choose *some* value as an input into structural models within economics, and to make power calculations during preregistration. Furthermore, the purpose of this or any meta-analysis is not to simply provide one number, but to demonstrate how such numbers vary given other factors that categorize studies using meta-regression techniques (e.g., figure 8). Stanley and Doucouliagos (2012) remind us that meta-regressions simply take a well-known and established technique to understand variation in many kinds of economic data and just apply it to the data our own profession generates.

Next, we turn to a sharper comparison between meta-analysis and narrative review. In a narrative review, there is no explicit attempt to canvas all studies based on stated criteria. Instead, an expert reviewer chooses studies that seem to be of especially high quality or pivot the scientific trajectory in a useful new direction. It is similar to an historical analysis of progress in a field. To relate the two methods, a narrative review is simply a meta-analysis with an altered subjective weighting system. The subjective weights are judgments by the reviewer of what findings readers should care the most about, and the reasons why.

One analogy is sports commentary. Meta-analysis is like a “play-by-play” analyst who describes every action on the field with an even tone. A narrative review is like a “color analyst” who picks out certain plays that are unusually important and explains why they are special. The color analyst *adds* the dramatic emphasis that the play-by-play analyst suppresses. The analogy should make clear why both kinds of commentary are useful; the two together are better than each one alone.²² A similar analogy is to the proud

²²Another useful analogy comes from performance evaluation in personnel economics. It is well-known that objective and subjective measures of performance can

divide in newspapers between the “news” side (meta-analysis) and “opinion” (narrative review).

A tricky, interesting question is what meta-analysis and narrative review can say about influential “breakthrough” papers. Narrative reviews often remark on how a particular study represents a breakthrough in using new methods or presents a surprising finding that should be prioritized to be studied further. Because meta-analysis is backward-looking, it is not ideally-equipped to identify useful breakthroughs. Because the narrative review is subjective, it can look forward and might do better.

An example is De Martino, Camerer, and Adolphs (2010). They found that two patients with damage to the amygdala area of the brain were not averse to losses at all. This is a tiny finding with a large standard error; it’s a teaspoon of water added to a swimming pool. Meta-regression is worthless because there is no power to detect “amygdala damage” versus “no amygdala damage” difference if the study were added to our corpus. But despite the tiny $n = 2$, it is a clue that could shed light on the fundamental mechanism underlying loss aversion, and hence deserves further study. Narrative reviews may miss some opportunities to amplify such interesting studies too, but meta-analysis will always neglect them.

We will conclude with one idea about how meta-analysis can guide future research. Meta-regression can actually pinpoint where studies are plentiful and where an additional study would have the greatest new effect on collective knowledge. A low standard error on a meta-regression coefficient means we do not need to learn more. A high standard error means that we do need to learn more.

From our dataset, studies (i) other than lab and field experiments; (ii) focusing on specific, nonuniversity student populations; (iii) on continents of South America, Africa and Oceania; (iv) involving rewards not expressed in monetary terms; (v) obtaining preferences in methods other than sequential binary methods; (vi) using utility functions other than CRRA with equal curvature for gains and losses; and (vii) with loss aversion specifications other than Kahneman and Tversky are the areas where we have uncovered the least data. Interesting new findings about the loss aversion parameter are ex ante more likely to be in those areas. The old cliché “we encourage future research in these areas” now has empirical backing.

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